

Gate E4 report — run_v1.21_r1 (Paper IV v1.21)

Verdict: **PASS** (**reused_verified_external_build**). Evidence: 20_EVIDENCE/E4/.

E4 — Documentary closure (method: **reused_verified_external_build**)

No Lean, Lake, leancchecker, compilation or negative-control compilation was run in this cycle. The E4 PASS of run_v1.2_r1 is reused after re-verification. Nothing was recompiled in this cycle.

Identity (E0): all 1097 files of run_v1.2_r1/40_PACKAGE/RUN_MANIFEST.json still match on disk. The sealed ZIP EXTERNAL_AUDIT_run_v1.2_r1.zip (sha256 886ed7f0...5e2b, 1098 members, CRC OK) matches its sidecar. Every item of the author's EXTERNAL_EVIDENCE_BINDING.json matches. The Lean source cut is unchanged (615/615).

Coverage re-verified by e4_reuse_check.py → E4_REUSE_CHECK.json, which does **not** use 10_LOGS/E4_main/SUMMARY.json (that file was overwritten by the AuditorChecks invocation, C-06):

- records.jsonl: 608 records and 608 distinct modules, all PASS. That is the 607 modules of the cut plus the auditor file AuditorChecks. All exit codes are 0 and no sorry flag is set.
- The source hashes in the records equal SOURCE_MANIFEST for all 607.
- E4_main_console.log final summary: modules_planned 607, pass 607, fail_or_blocked [], finished 2026-09-30T21:33:08Z. All 19 targets are PASS, the console ends with "EXIT 0", and there are 607 "PASS i 607" lines.
- 608 module logs; case-insensitive "sorry" appears in none. The four log sets (auditor main 608, annex 50, author segment 1 422, segment 2 186) were rescanned with the corrected detector: 0 hits.
- 19 target logs: 461 records under the author rule, 314 of them #print axioms, with no non-standard axiom.
- 224 #check lines in ReleaseExportCheck.lean; its log has 0 error lines.
- Annex: 50 records, 50 PASS. BoundedCliquesGap audit line: 633 declarations with standard axioms; the five cited theorems depend on exactly [propext, Classical.choice, Quot.sound].
- olean_vs_author.json: 607/607 byte-identical. AuditorChecks.log: "AUDITOR CONE SUMMARY: 50 declarations; problems=0". cache_diff.json: shared cache unchanged.
- Types and axioms of the headline declarations are as recorded in run_v1.2_r1 E3/E4 and are unchanged, because the source is unchanged.

Limits kept: same machine; shared pinned third-party oleans, not rebuilt; no independent kernel replay.

E4 verdict: PASS (**reused_verified_external_build**).

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